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KALIMATI, KATHMANDU



Minor Project Mid Defense Report

On

Tribal: Find Your Tribe

[Code Number: ENCT354]

BY

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TO

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Abstract

Finding compatible roommates and meeting like-minded people in a new city is a problem that many people face today. Most of the existing platforms either focus only on roommate searching or only on social activities. There is no single platform that can help users find roommates, join activities, and build a trusted community all at once. TRIBAL is a mobile application built using Flutter that tries to solve this problem. The application allows users to find compatible roommates based on their lifestyle preferences, find activity partners, create and join local events, and chat with other users. The platform also includes a compatibility scoring algorithm that matches users based on factors like budget, sleep schedule, cleanliness habits, pet preferences, and other lifestyle-related questions collected during onboarding. The backend of the application is built using Django and uses PostgreSQL as the database. Authentication is handled using JWT tokens and real-time chat is being implemented using Django Channels with WebSockets.

Keywords: Roommate Matching, Compatibility Scoring, Activity Partner Discovery, Mobile Application, Flutter, Django, PostgreSQL, Real-Time Chat, Community Building, Lifestyle Preferences

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Abbreviations

API	Application Programming Interface
Dart	Dart Programming Language
Django	Django Web Framework
Figma	Figma Design Tool
Flutter	Flutter UI Framework
Git	Git Version Control System
GitHub	GitHub Code Hosting Platform
JWT	JSON Web Token
MVC	Model-View-Controller
PostgreSQL	PostgreSQL Database
Postman	Postman API Testing Tool
Python	Python Programming Language
REST	Representational State Transfer
UI	User Interface
UML	Unified Modeling Language
UX	User Experience
VS Code	Visual Studio Code

Chapter 1 Introduction

1.1 Background Theory

In recent years, the increasing number of students and working professionals moving to new cities has created a growing need for platforms that help users find compatible roommates and build social connections. Existing social media and roommate-finding applications either lack compatibility-based matching or focus only on basic profiles, leaving a gap in the market.

TRIBAL is a cross-platform mobile application developed using Flutter and Django to address this problem. It enables users to create profiles, answer lifestyle-based onboarding questions, and find compatible roommates and activity partners through a weighted compatibility scoring algorithm. The application also supports event creation, participation, and real-time chat between matched users.

The weighted compatibility scoring approach was chosen over machine learning because it is transparent, easier to implement within the scope of a minor project, and provides meaningful matches based on users' preferences and interests.

1.2 Problem Statements

People who want to find others with shared interests often have no proper platform to do so. General social media platforms are too broad and not designed for this kind of need. Activity-specific platforms like Meetup are not widely used in Nepal and do not support roommate finding at all.

The main problems that TRIBAL tries to address are:

- Finding compatible roommates and building a social circle is challenging for students and young professionals who move to new cities, as they often rely on informal methods such as friends or social media groups.
- Existing platforms do not provide a single solution for roommate matching, activity partner discovery, and community building, while also overlooking important lifestyle compatibility factors.

- TRIBAL addresses these challenges by offering a unified platform for roommate matching based on compatibility, activity partner discovery, event participation, and secure in-app chat.

TRIBAL aims to solve these problems by providing a single mobile platform where users can find compatible roommates, discover activity partners, join events, and chat with matches in a safe environment.

1.3 Objectives

The main objectives of this project are:

- To develop a mobile platform that helps users find compatible roommates by evaluating lifestyle preferences through a compatibility scoring system.
- To provide a platform where users can create, browse, and join local events to encourage community engagement.
- To facilitate secure one-to-one communication between matched users through an in-app real-time chat system.

1.4 Scope and Applications

The scope of this project as a minor project includes the following features:

- Develop a cross-platform Flutter application that enables users to find compatible roommates and activity partners through secure authentication, profile management, and preference-based matching.
- Provide essential social features, including real-time chat, event creation and participation, and interest-based activity discovery to help users build meaningful connections.
- Address the challenges of finding roommates and social networks, particularly for students and newcomers in Nepal, while serving as a scalable prototype for future enhancements such as verified profiles and location-based recommendations.

For the purpose of this minor project, the focus is on completing the core features including the compatibility matching system, real-time chat, and event module. The

application is not intended for commercial deployment at this stage, but it demonstrates a working prototype of the complete system.

1.5 Literature Review

1.5.1 Recommendation and Matching Systems

Recommendation systems are used to suggest suitable people, items, or services based on user preferences and behavior. In social applications, they help users find people with similar interests and lifestyles. TRIBAL uses this concept to recommend compatible roommates and activity partners. Since a large dataset is not available for this minor project, a rule-based weighted matching system is used instead of machine learning.

1.5.2 Roommate Matching Systems

Existing roommate matching systems help users find roommates based on preferences such as budget, food habits, cleanliness, sleep schedule, and lifestyle. These systems show that roommate compatibility depends on more than just room availability. However, most roommate matching systems only focus on finding rooms or roommates and do not include activity discovery, events, or community-building features.

1.5.3 Event and Activity Recommendation

Event recommendation systems help users discover activities based on interests, location, or user behavior. These systems are useful for connecting people with similar hobbies in real life. TRIBAL follows this idea by allowing users to create, browse, and join local events. However, instead of using complex location tracking, TRIBAL uses interest-based event discovery, which is simpler and more suitable for this project.

1.5.4 Review of Existing Platforms

Platforms like Roomster, Roomi, and SpareRoom focus mainly on roommate searching, while Meetup and Bumble BFF focus on social connection and activity discovery. These platforms solve only one part of the problem. In Nepal, many users still depend on Facebook groups or personal contacts, which lack proper compatibility checking, verification, and structured matching.

1.5.5 Research Gap

From the reviewed systems, it can be seen that existing platforms usually focus separately on roommate matching, friend discovery, or event participation. There is a lack of a single mobile application that combines compatible roommate matching, activity partner discovery, local events, and secure chat. TRIBAL addresses this gap by using a weighted compatibility scoring algorithm based on lifestyle and interest-related onboarding answers.

Chapter 2 Methodology

2.1 Process Model

The team is following the Agile process model for the development of TRIBAL. Agile was chosen because it allows the project to be broken down into small, manageable parts called sprints, and each sprint produces a working piece of the application. This is useful for a minor project because requirements can be refined as the team learns more, and progress can be shown at regular intervals like this mid-term defense. The project has been divided into the following sprints:

- Sprint 1: Project setup, requirement analysis, and UI/UX wireframing in Figma.
- Sprint 2: Flutter onboarding flow, authentication screens (Login and Signup), and post-signup profile completion screens.
- Sprint 3: Django backend setup, PostgreSQL database design, and JWT-based authentication APIs.
- Sprint 4: Design and implementation of the roommate compatibility scoring algorithm (current sprint).
- Sprint 5: Real-time chat system using Django Channels and WebSockets.
- Sprint 6: Event creation and joining module.
- Sprint 7: Integration testing between the Flutter frontend and Django backend.
- Sprint 8: Final testing, bug fixing, and documentation.

At the time of this report, the team has completed Sprints 1 to 4 and has started work on Sprint 5.

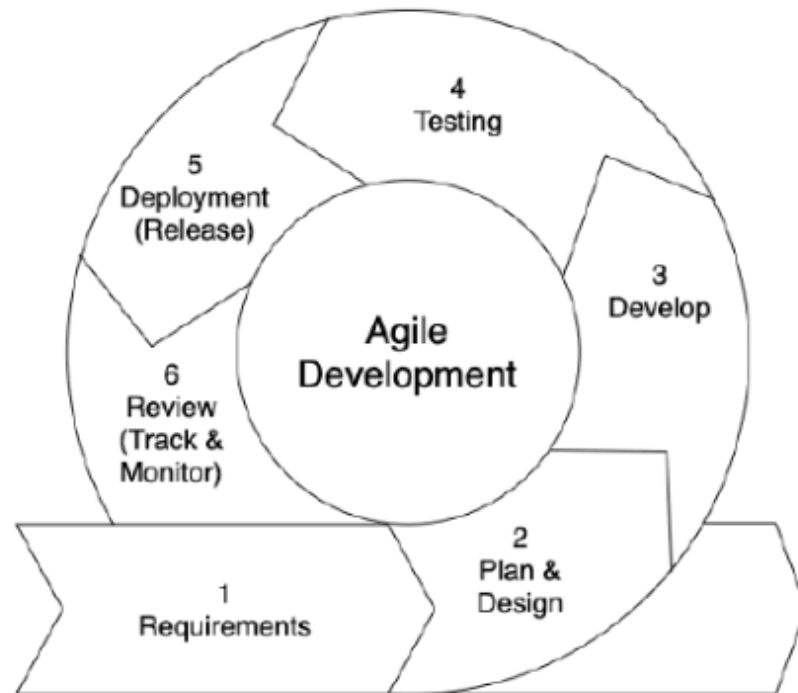


Figure 2.1: Agile Process Model

Figure ?? shows the Agile development cycle that the team is following for this project. The diagram is a circular flow with six stages: Requirements, Plan & Design, Develop, Testing, Deployment (Release), and Review (Track & Monitor), all revolving around a central label that says “Agile Development”. The circular arrangement shows that after each review cycle, the team loops back to plan the next set of requirements. This model fits our project well because we are working in short sprints, and after each sprint we review what was done and plan the next one based on what is remaining.

2.2 System Block Diagram

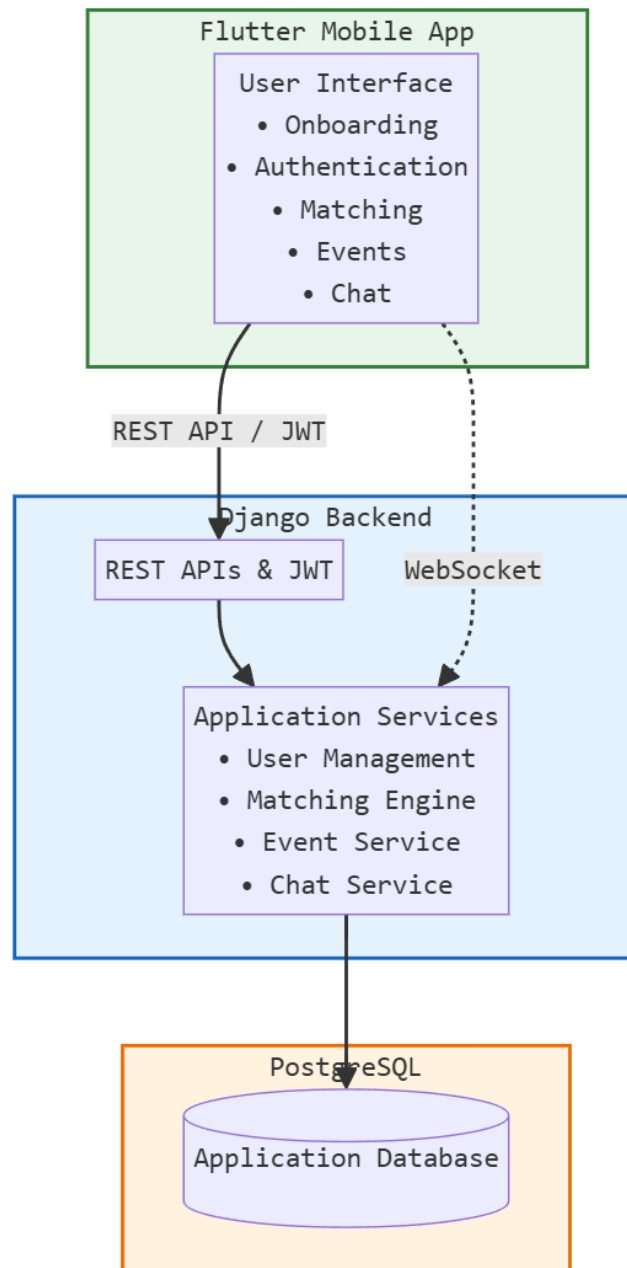


Figure 2.2: System Block Diagram of TRIBAL

Figure ?? shows the overall system architecture of TRIBAL divided into three layers. At the top is the Flutter Mobile App, which contains the User Interface with five sections: Onboarding, Authentication, Matching, Events, and Chat. Below it is the Django Backend, which handles two kinds of communication: regular REST API requests and JWT-based authentication on the left, and WebSocket connections for real-time chat on the right. Inside the Django layer, an Application Services block handles four services: User Management, Matching Engine, Event Service, and Chat Service. At the bottom is

PostgreSQL, represented by a database cylinder labelled Application Database. A solid arrow connects the Flutter app to the Django backend via REST API/JWT, and a dashed arrow shows the WebSocket connection used for the chat feature. This layered structure makes it easy to understand how each part of the system is responsible for a specific job.

2.3 Algorithm

2.3.1 Roommate Compatibility Scoring Algorithm

The main algorithm used in TRIBAL is the Roommate Compatibility Scoring Algorithm. It compares the onboarding responses of two users and calculates a compatibility score between 0 and 100. The algorithm is rule-based and uses weighted scoring because it is simple, transparent, and suitable for a minor project prototype. The algorithm takes two user profiles as input: the current user and a candidate roommate. It compares their lifestyle preferences such as cleanliness, budget, sleep schedule, noise level, smoking, guests, interests, study habit, food preference, and drinking habit. Input Current user profile and candidate roommate profile containing: Cleanliness preference Budget range Sleep schedule Noise preference Smoking preference Guest preference Interests Study habit Food preference Drinking preference Output A final compatibility score between 0 and 100.

Table 3.1 below shows the weight distribution used in the scoring algorithm.

Factor	Weight (%)
Cleanliness	15
Budget	15
Sleep Schedule	12
Noise Level	10
Smoking	10
Guests	8
Interests	10
Study Habit	8
Food Preference	7
Drinking	5

Table 2.1: Compatibility Scoring Weights

The general formula used to calculate the final compatibility score is:

$$\text{Final Score} = \sum (\text{Factor Match Value} \times \text{Factor Weight}) \quad (2.1)$$

Where Factor Match Value is a number between 0 and 1 that represents how closely two users' answers match for a particular factor, and Factor Weight is the percentage weight assigned to that factor as shown in Table 3.1. For factors with a small number of fixed options, such as smoking or sleep schedule, the match value is calculated using a simple rule: if both users selected the same option, the match value is 1; if their answers are close but not the same (for example, "early riser" and "no preference"), a partial match value such as 0.5 is given; and if the answers directly conflict (for example, "smoker" matched with "non-smoker, no smoking allowed"), the match value is set to 0. For factors that involve multiple selections, such as interests, the match value is calculated as the number of common interests divided by the total number of unique interests selected by both users combined. This gives a percentage overlap between the two users' interests. For numeric factors like budget, the match value decreases as the difference between the two users' stated budgets increases. If the difference is within an acceptable range, the match value stays close to 1. If the difference is very large, the match value moves closer to 0.

2.3.2 Deal Breakers

Apart from the weighted scoring, the algorithm also includes a deal breaker check. Some lifestyle differences are considered serious enough that they should not simply lower the score a little, but should instead strongly affect or override the final match. For example, if one user is a heavy smoker and the other has explicitly said they cannot live with a smoker at all, this is treated as a deal breaker rather than just a normal mismatch. Table 3.2 lists the deal breaker rules currently used in the algorithm.

Condition	Effect on Score
Smoking conflict (smoker vs. strictly non-smoking)	Score capped at a low maximum
Extreme budget mismatch (outside acceptable range)	Score capped at a low maximum
Opposite cleanliness extremes (very messy vs. very strict)	Score significantly reduced

Table 2.2: Deal Breaker Rules

When a deal breaker condition is detected, the algorithm does not let the final score go above a fixed low ceiling, even if all other factors match well. This makes sure that users are not shown as highly compatible with someone they have a fundamental lifestyle conflict with.

2.3.3 Algorithm Steps

article

algorithm algpseudocode

Algorithm 1: Roommate Compatibility Scoring

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1 [1]
2 Current user profile, candidate roommate profiles Ranked list of compatible
  roommates
3 Get current user's onboarding responses Fetch candidate roommate profiles from
  the database
4 for each candidate profile do
5   total_score ← 0
6   Calculate cleanliness_score Calculate budget_score Calculate
     sleep_schedule_score Calculate noise_level_score Calculate smoking_score
     Calculate guest_score Calculate interest_score Calculate study_habit_score
     Calculate food_preference_score Calculate drinking_score
7   total_score ← cleanliness_score + budget_score + sleep_schedule_score +
     noise_level_score + smoking_score + guest_score + interest_score +
     study_habit_score + food_preference_score + drinking_score
8   if deal breaker condition exists then
9     Apply score cap or reduce total_score
10  final_score ← round(total_score) Store candidate with final_score
11  Sort candidates in descending order of final_score return ranked list of
     compatible roommates

```
